

Section 1: Executive Summary

FitLife is a Health & Fitness mobile application with approximately 50,000 monthly active users and an average rating near 4.1. Despite a strong foundation, growth has plateaued and Day-7 retention remains relatively weak at approximately 22%. Review analysis indicates that while users value workout tracking, calorie logging, and interface simplicity, recurring complaints involve missing offline access, technical instability, sync problems, and perceived subscription pricing concerns.

Across Milestones 1 and 2, VADER sentiment analysis, TF-IDF keyword extraction, and feature tagging were applied to 1,561 reviews across seven comparable Health & Fitness applications. Milestone 3 translated these insights into a prioritized feature backlog using the RICE framework. “Rest timer between sets” emerged as the top quick-win feature (RICE 4,646), followed by Dark mode UI themes and Advanced nutrition/meal planning.

Milestone 4 proposes a statistically rigorous A/B test to evaluate the Rest timer’s impact on Day-7 retention, targeting an uplift from 22% to 27%. Additionally, it consolidates insights into a Tableau dashboard integrating sentiment analysis, feature sentiment, RICE prioritization, and experiment design to guide continuous product improvement.

Section 2: User Sentiment Insights

Overall sentiment analysis reveals a mildly positive average polarity aligned with an average rating of 4.1/5. While 4- and 5-star reviews dominate, a meaningful proportion of 1- and 2-star reviews highlight dissatisfaction. Negative reviews tend to be longer and more detailed, providing actionable product insights.

Feature-level analysis shows Workout Tracking and Calorie Nutrition categories with the highest positive sentiment. UI/UX also reflects favorable perception. In contrast, Offline Mode shows the lowest sentiment polarity, followed by Bug/Technical issues and Sync/Integration problems. Premium pricing concerns also appear frequently in negative feedback.

Feature request extraction indicates strong demand for Offline Mode, Wearable Integration, Meal Planning, and Social Features. Emotional analysis suggests trust and joy dominate overall sentiment, while frustration clusters around connectivity and reliability issues.

Section 3: Feature Prioritization

Using the RICE framework ($\text{Reach} \times \text{Impact} \times \text{Confidence} \div \text{Effort}$), 12 candidate features were evaluated. The top five ranked features are:

1. Rest Timer Between Sets (RICE: 4,646)
2. Dark Mode UI Themes (RICE: 2,636)
3. Advanced Nutrition/Meal Planning (RICE: 527)
4. Voice-Guided Workouts (RICE: 290)
5. Video Exercise Demonstrations (RICE: 290)

Rest Timer ranks highest due to strong implied demand, direct enhancement of core workout functionality, and low implementation effort.

Section 4: A/B Test Plan – Rest Timer

FitLife Product Strategy – Milestone 4

Hypothesis: Introducing a Rest Timer will increase Day-7 retention from 22% to 27%.

Design:

- 50/50 randomized split (Control vs Treatment)
- 1,076 users per group (2,152 total sample)
- 21–28 day test duration
- Primary metric: Day-7 retention
- Secondary metrics: session frequency, engagement, session duration
- Guardrails: crash rate, load time, uninstall rate

Statistical method: Two-proportion z-test ($\alpha = 0.05$, power = 80%).

Launch criteria: $p < 0.05$ AND ≥ 5 percentage point uplift.

Section 5: Product Roadmap (6 Months)

Q1:

Month 1 – Rest Timer development and experiment

Month 2 – Dark Mode implementation

Month 3 – Begin Advanced Nutrition development

Q2:

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Months 4–5 – Complete Advanced Nutrition

Month 6 – Voice Guidance and Video Demonstrations rollout

Expected impact:

- Day-7 retention: 22% → 27%+
- Rating: 4.1 → 4.3–4.4
- MAU: 50K → 60K+

Section 6: Implementation & Monitoring

Implementation will follow a Design → Development → QA → Experiment → Launch framework. Weekly reviews will track progress and experiment results. KPIs include WAU, MAU, retention cohorts, feature adoption, app rating, sentiment trends, and competitive benchmarking.

Six-month targets include raising average rating to 4.3–4.4+, increasing MAU beyond 60K, improving Day-7 retention toward 30%, and significantly reducing negative sentiment around offline access and technical issues.